

Chapter 14

Ethical Relationships, Legal Frameworks, and Agreements



Randolph Hall

14.1 Introduction

University–industry (UI) relationships are executed within the framework of agreements that articulate the terms and conditions of their work together. Agreements capture both the specifics of particular relationships—work products, schedule, payments, start and end dates, and so on—and general terms, such as legal obligations, procedures for protecting Intellectual Property (IP), liabilities, and indemnification. Beyond mutual agreement to follow the law, companies and universities seek to ensure that the relationship provides benefits commensurate with costs and risks, and is consistent with their mission and values.

UI agreements, such as sponsored research contracts, are often challenging to negotiate because, as introduced in Chap. 1, each party values different things. Though universities and companies both look for creativity, novelty, and impact, as well as workforce development, economic growth, and IP, they prioritize different outcomes. A company seeking commercialization may worry that a university partner will be too slow or too careless in protecting IP. A university may worry that a company will impede the publication of research or degree progress for a doctoral student. All of these concerns are important and valid, and should be captured in the agreement in ways that keep the relationship on track to meet everyone's goals. Further, it is important for the individual participants—staff, students, faculty, management—to be aware of, and commit to, the stated obligations.

Ethical conduct is always important. Honesty, meeting commitments, “do no harm,” and respect for others are examples of ethical principles that universities and

R. Hall (✉)

Epstein Department of Industrial and Systems Engineering, University of Southern California, Los Angeles, CA, USA

e-mail: rwhall@usc.edu

Mechanics			
Foundations		Avenues	
Alignment	Organization	Academic Health System	External Relations
Barriers	Partnerships	Alumni	Fundraising
Engagement	Roadmaps	Clinical Research	Gaining Support
Grand Challenges	Stakeholders	Consortia	Proof of Concept
Innovation	Strategy	Co-operative Education	Research Parks
Interfaces	Values	Corporate Venture	Skill Development
		Economic Development	Sponsored Research
		Entrepreneurship	Startups
		Executive Education	Venture Capital
		Agreements	Technology Platforms
		Commercialization	Training
		Conflicts of Interest	Material Transfer
		Copyrights	Membership
		Data Security/Sharing	Confidentiality
		Intellectual Property	Commitments
		International Partnerships	Export Control
		Licensing	Transparency
		Patents	Privacy
		Policies	Responsible Research
		Regulations	Publicity

Fig. 14.1 Mechanics of successful UI relationships

companies can easily agree to. In fact, it would be hard for any two parties to execute an agreement without these basic commitments. Other values may be less universal, such as academic freedom, independence, ownership, and transparency. Parties may spend significant time negotiating terms and conditions for which their values, and explicit policies, differ.

This chapter introduces the book section on the mechanics of UI relationships, focusing on the topics highlighted in the outer section of Fig. 14.1. Specifically, in this chapter, UI relationships are examined in the context of agreements, which capture the details of achieving strategic, ethical, and legal objectives, in accordance with organizational values. The issues in Fig. 14.1 need to be resolved among the parties in a UI agreement when moving from informal relationships to formalized engagements. Agreements must define the specifics of how the parties will collaborate toward mutual goals.

14.2 Challenges and Opportunities

Formal UI engagements—those executed through signed agreements between organizations—seldom begin without prior informal relationships. Unlike the National Institutes of Health (NIH) and National Science Foundation (NSF) granting processes, where agencies post detailed solicitations for proposals, and proposals are reviewed and prioritized by committee, UI relationships are catalyzed by conversations between company and university researchers with similar interests. The starting point may be professional conferences, organized site visits, graduated students employed in industry, students holding internships, or just a realization that related research and development are occurring in both places. Once the informal relationship is established, conversations may begin as to how shared interests can move into a formalized engagement, such as industry-sponsored research, data or resource sharing, exchange programs, clinical trials, executive education, or any of the other avenues described in prior chapters.

Through formal engagements, universities and industries have an opportunity to collectively apply their resources—money, talent, equipment, laboratories, data, and so on—toward their mutual goals. Agreements can lead to expanded and

multidimensional relationships, involving more people, targeted outcomes, and pathways to commercialization. Formal engagements also have the opportunity to articulate expectations and requirements so that the partners know their commitments to each other, helping avoid misunderstandings.

The challenge of formal engagements is that agreements take time to negotiate, during which progress might be impeded. Terms, conditions, and payment systems are not standardized. Industry agreements vary from company to company, and even sometimes within a company. It takes time to create the mutual understanding that produces an executed agreement. For these reasons, companies and universities must learn about each other's expectations, needs, and options. With these points in mind, the following chapter sections outline some of the key issues.

14.3 Options, Solutions, and Impact

Universities and companies negotiate institutional agreements through strategic, legal, and ethical lenses, considering costs, benefits, and risks. As they negotiate, each party must assess how they will successfully fulfill their obligations under the agreement and how they can ultimately achieve their organizational goals, such as commercialization, profitability, reputation enhancement, public service, knowledge creation, and student success. Institutional agreements come in many forms, such as those described in Table 14.1. In some cases, an initial agreement, such as a confidentiality agreement, is a precursor to a more substantial sponsored research agreement. Some agreements serve narrow and tightly specified purposes, such as material transfer. Other agreements may be more expansive and strategic, such as an agreement to create a joint venture.

Other than institutional agreements, university faculty, staff, and students sometimes individually enter into work-for-hire consulting relationships with companies. Typically, the university is not a party to consulting agreements (i.e., they are not signatories). However, work-for-hire consulting relationships must comply with university policies, particularly with respect to conflicts of interest and the use of university resources. Lately, it has also become more common for faculty to be jointly employed with companies on a split-time basis. Such arrangements must be carefully reviewed and approved by both the company and the university, whether or not they enter into an agreement with each other.

Any formal engagement between a university and a company should begin with an exploration of the type of agreement that fits the purpose. For instance, companies may prefer gift relationships for their simplicity but must be aware that they can receive no rights in exchange for their gift. Others will prefer sponsored research agreements because they will have greater oversight over the research project along with a clearer pathway to commercialization. No matter what type is selected, an agreement binds the organizations to its terms and conditions, which are enforceable under the law. For this reason, negotiation should be preceded by answering five key questions:

Table 14.1 Example types of UI institutional agreements

Name	Description
Clinical trials	Testing of a technology, pharmaceutical or procedure in a controlled environment with human or animal subjects following an approved protocol, for cost reimbursement.
Confidentiality	Specification of information to be shared and held in confidence, for a specified use, often for the purpose of proposal development and typically without payment.
Data Use	Agreement to share data between organizations, often research studies, with designated limitations and responsibilities for the recipient, typically without payment.
Gift	An irrevocable payment for a designated purpose, such as research, without promised deliverables or IP rights.
Joint Venture	Partnerships between organizations to create a new jointly controlled and owned entity, such as a commercialization venture.
Material Transfer	Agreement to provide research materials—such as biomaterials, compounds, or animals—from one party to another for specified uses, often with payment made by the recipient.
Membership	Payment to participate in a university center, such as a National Science Foundation Engineering Research Center, with specified rights, including input on the work of the center and shared access to center products.
Service	Utilization of unique resources, such as laboratory equipment, to perform a specified task for cost reimbursement. Service agreements are not research because the sponsor defines the proposed work.
Sponsored research	A contracted relationship where a company funds a research project with a specified schedule, promised deliverables, and IP rights.
Teaming	Agreement to work together in pursuit of funding from a third party, typically a government agency.
Technology licensing	Provision of rights to IP under specified terms and prices, for further development and commercialization.

1. How does the relationship align with the organization's strategic goals, what will it accomplish and does it provide benefits that justify the costs and risks?
2. What agreement type best fits the strategic goals?
3. How will the parties ensure that the agreement complies with the law and regulations, their internal policies and organizational values?
4. Is each organization capable of performing as promised once the agreement is executed?
5. What activities will occur at the end of the agreement to achieve the relationship's aims?

With respect to the first question, from the company perspective, access to talent, innovative ideas, and new IP are common motivators. For universities, common motivators are research support, understanding of applications and needs, and pathways to public benefit, all of which are well aligned. On the other hand, some companies may seek out universities for evaluations that help promote their products.

Such promotional goals are likely not aligned with university policy, which values independence. Without alignment, the university is likely to reject the relationship.

No one person has sufficient insight to answer all five questions. Communication inside each organization—including performers (e.g., principal investigators), contracting officers, compliance officers, technology transfer officers, and senior leadership—is crucial. Each person contributes an important perspective that must be shared with others to arrive at a negotiated agreement that fulfills strategic, legal, and ethical goals. Prior chapters have addressed strategic considerations. The following sections of this chapter discuss legal and ethical considerations.

14.3.1 Legal

As companies and universities enter agreements, they may discover that they are exposed to laws and regulations they do not normally encounter. Sometimes, the laws can inhibit normal ways of doing business, and, as a result, they must find ways to adapt or forego the relationship. One reason is that government regulations differ between typical university research, normally classified as fundamental, and industry research and development, which might carry restrictions. Another reason is that company employees working on internal development projects are not subject to the same rules as universities aiming to publish research results for outside consumption. A third reason is that universities, unlike commercial contractors, do not normally operate on a work-for-hire basis and, as a result, retain ownership of the IP they produce. Here are a few examples of relevant laws and regulations.

Financial Accounting and Payment Research budgets at US universities follow accounting principles governed by the Office of Management and Budget's Uniform Guidance for Federal Awards, which became effective in 2014. Key features include separating direct costs (those directly attributed to a funded project) from indirect costs (facility and administrative costs that are not directly attributed). Employee wages are a direct cost calculated from actual salaries and months of effort (e.g., number of months multiplied by a percentage effort per month), but fringe benefit costs are calculated as a percentage of wages. Unlike a typical agreement with a law or consulting firm, the budget reveals the exact salary of each employee, as shown in the budget.

Both fringe benefit rates and indirect cost rates are subject to audit, but the administrative portion of the indirect cost is capped at 26%. For consistency, universities rarely construct research budgets based on hours of effort, and they rarely utilize fully loaded hourly rates, though they would be permitted to do so for industry-sponsored research. They also typically do not charge fees on top of audited costs. Given that administrative costs are capped, the accounting system, by design, requires universities to subsidize research projects. Industry-sponsored clinical trials (Chap. 11) is an exception to these practices in that many site costs are

charged as direct expenses, indirect rates are lower than typical federal rates, and clinical costs are determined according to “Fair Market Value” determinations.

Export Control and International Considerations Unlike government-sponsored fundamental research in the United States, industry-sponsored applied research may be subject to International Traffic in Arms Regulations (ITAR) and Export Administration Regulations (EAR), which govern the movement of critical technologies out of the country. Compliance with such regulations is difficult for universities, which offer an open culture, including many foreign citizens, including citizens of countries subject to sanctions of the Office of Foreign Assets Control (e.g., Iran, a source of many graduate students). US universities also have many students from China, which is subject to the prohibition of military exports. Regulations control both physical exports and “deemed exports,” through which individuals may gain access to technical data by participating in a research lab. Universities are very reluctant to accept such restrictions for two reasons: first, as a matter of principle that they do not want to prohibit students from participating in research and, second, because it is difficult for them to assure compliance.

IP Protection The Bayh–Dole Act of 1980 conveyed ownership of IP resulting from federally funded research to the businesses and universities that created it. A key provision is that universities and other nonprofits must distribute a portion of royalties to inventors, thus treating university researchers differently from industry researchers, who typically do not directly share in royalties. A consequence is that university IP policies explicitly recognize the rights of inventors, which must be respected in any UI agreement.

Privacy Rules The privacy of certain types of personal information is protected by US government regulations. For example, access to personal health information is strictly regulated by Health Insurance Portability and Accountability Act (HIPAA) and access to personal student information is strictly regulated by the Family Educational Rights and Privacy Act (FERPA). The European Union’s General Data Protection Regulation (GDPR) protects the privacy of personal data, including a personal right to data erasure. The U.S. Privacy Act of 1974 protects personal data held by the US government, and is sometimes applied to federally sponsored research. These laws are examples of how companies and universities must be cautious about the legality of sharing personal data with each other, such as data collected by social media companies or any other Internet-focused enterprise.

Open Access and Transparency Regulations and policies require transparency and access for many aspects of federally funded research. As directed by Alondra Nelson of the U.S. Office for Science Technology and Policy in 2022, federal agencies that fund research and development are obligated to establish public access policies for peer-reviewed scholarly publications and scientific data (Nelson, 2022). While these policies do not directly apply to privately sponsored research, if a research project is jointly sponsored with the US government, open access may be

required. The US government also requires clinical trials to be registered within 21 days of enrolling their first subject. The Sunshine Act requires manufacturers of drugs and medical devices to report all payments in excess of \$10 to physicians and teaching hospitals, which are posted in the Open Payments database. Universities can also be required to disclose information to the public under the U.S. Freedom of Information Act (FOIA) for federally funded research. Public universities may additionally be subject to state public information acts, requiring disclosure of information on research relationships with private industry.

Human Subject Protection The “Common Rule” (OHRP, 2018) regulating human subject research in the United States requires review and approval of research protocols by Institutional Review Boards (IRBs), along with informed consent by research subjects. The common rule defines research as “systematic investigation, including research development, testing, and evaluation, designed to develop or contribute to generalizable knowledge.” Private companies conducting human subject R&D with their own funds are not required to follow the common rule. However, universities conducting human subject research with company funding will utilize their IRBs for review and utilize informed consent. As a consequence, a UI partnership may subject research to regulations that companies are not accustomed to follow.

To summarize, universities must be cognizant of the laws and regulations that apply to industry-sponsored research, and companies must be cognizant of the laws and regulations that bind universities. All applicable laws and regulations must be respected in the agreements that they negotiate.

14.3.2 *Ethics, Values, and Policies*

Universities generally ascribe to a set of academic values that can be traced to the early twentieth century, including *Academic Freedom*, which protects faculty freedom in their teaching, research, intramural speech, and extramural speech. Such rights are typically expressed in handbooks, mission statements, and value statements.

With respect to research, the Association of University Professors (AAUP, 2025) states that academic freedom encompasses “freedom to explore all avenues of scholarship, research, and creative expression and to publish the results of such work.” Such freedom generally encompasses the right to conduct research sponsored by all types of companies, even those that some in the academic community might view as unsavory. On the other hand, academic freedom, unlike constitutionally protected freedom of speech, is limited by the duty of faculty to truthful research. Faculty can be fired when they commit *Research Misconduct*, which includes plagiarism, fabrication, and falsification, as defined by the U.S. Office of Research Integrity and university policies. Short of deliberate misconduct, faculty

and students are expected to meet quality standards in their research as part of their general obligation for *Responsible Conduct of Research (RCR)*, a subject that includes ethics (including human and animal subject protection and safety), data management, experimental design, and reporting.

Universities often require students and faculty to complete RCR training as a condition for conducting research. Nevertheless, quality control is much less systematic in universities than in industry, owing to a culture that values creative independence, along with a shortage of institutional controls and infrastructure supporting the conduct of research. In addition, because the people doing the research are often students with minimal experience, the research process can be difficult to manage at industry standards. University contracting offices recognize this point in that they are very reluctant to assure or guarantee research quality.

Academic freedom implies that universities do not regulate or control what faculty and students publish or present, and they typically do not enter into agreements that grant research sponsors a right to control research products. However, universities will commonly provide sponsors with the right to review publications to ensure that confidential information is not disclosed and that IP is adequately protected, which may require that publication is postponed. Some universities, such as the University of Southern California, nevertheless accept publication restrictions (including classified research) when the research is deemed to serve national security interests, but only when reviewed and approved by a faculty committee.

Though faculty and students enjoy academic freedom, they must also comply with standards of *Attribution*, as set forth by disciplines, journals, and universities. In general, any person who made a substantive contribution to research should be named as an author or creator of the research product, which might include data, software, and interactive media, as well as articles. Persons who did not make a substantive contribution should not be named. An implication is that “ghost authors” (e.g., an unnamed author) are unacceptable. Orders of authorship must also reflect the relative contributions of the research team, though the standards for who is named first and last vary by discipline. Of late, the “CRediT Taxonomy” is a standard used by many journals through which all authors must precisely specify what they contributed to a piece of research.

Universities must recognize that academic freedom is not an inherent right in private industry, where publications typically require managerial approval, and likewise, companies must recognize that freedom to publish research is an inherent right for students and faculty. Agreements must be structured in ways that respect these differences while achieving the strategic goals of the relationship.

Another right typically conveyed to faculty is the freedom to work as paid consultants for industry, within a limit of 1 day/week. In that capacity, faculty can individually choose to accept confidentiality terms that restrict their right to publication. The right to consult is limited by *Conflict of Interest in Research* and *Conflict of Commitment* policies, whose considerations may include whether the relationship (1) causes the faculty member to sacrifice objectivity in research; (2) jeopardizes the independence of supervised students; (3) inappropriately utilizes university resources; (4) violates the primary duty of the faculty member to serve the

university; (5) conflicts with sponsored research agreements, including IP provisions; or (6) exploits the name of the university as an implied endorsement for a company. Conflict of interest policies at universities require disclosure, review, and approval, often by committee. Some types of relationships are prohibited by policy, such as paid promotional speaking by faculty on behalf of pharmaceutical companies, or paid consulting for research that is better done under a sponsored research agreement with the university. Conflict of interest policies must be consistent with the regulations of federal research sponsors, such as those established by the Department of Health and Human Services.

In addition to policies regulating individual conflict of interest, universities also establish policies on *Institutional Conflict of Interest*, which address circumstances where bias might be introduced when the university—not just the individual—stands to benefit financially. An example is clinical trials for pharmaceuticals for which the university owns the underlying IP. In such instances, the university may allow an early phase I trial in its clinics but not permit later phases. It may alternatively elect to divest its financial interest in the IP.

IP policy is critical for any research agreement as it defines protection, rights, responsibilities, and commercialization pathways. In the United States, the IP policy must comply with the Bayh–Dole Act, as previously mentioned, which conveys IP ownership to the employee’s institution for federally sponsored research. Though, technically, industry-sponsored research does not need to follow Bayh–Dole, university policies tend to apply uniform rules independent of the sponsor, for consistency and clarity among researchers and staff. Internationally, universities tend to be like the United States, whereby institutions own IP. Still, in some cases the creators own the IP (Waterloo University, mentioned in an earlier chapter, is an example). Under standard policy, creators are required to disclose their inventions and other IP to their university, which then decides whether and how the IP is protected and commercialized. While the process may seem straightforward, the technology transfer office often needs to balance conflicting objectives and interests:

- Was the IP created through multiple channels of support, perhaps a combination of federal and industry funds, for which sponsored research agreements must be resolved relative to each other?
- If the IP was co-created with a company, who is responsible for its protection and how is income distributed?
- Were student and faculty creators informed and did they consent to any sponsored research agreement terms that conveyed rights to an industry sponsor?
- Is the best channel for commercialization through an established company or via a startup, for which the creators hold personal financial interests? Along with this, if a faculty member has a startup, is he or she complying with the rights of the industry sponsor?

These questions revolve around a key ethical principle: disclosure and informed consent. In addition, university technology transfer offices try to respect the priorities set by faculty principal investigators as to how IP is protected and commercialized. In some cases, this may mean placing IP in the public domain, as in open-source

software or freely available technology. Doing so must naturally be consistent with any research agreement with companies, some of whom may prefer public domain as the best commercialization option. The next chapter continues the discussion of IP in depth.

Research relies on and creates valuable data, which is essential to understanding and conducting experiments, protecting IP, validating research quality, replicating experiments, and supporting future research and development. Universities and industry are alike in that the institution owns research data. Universities do so in part because their sponsors expect them to control data. Continuity of research, particularly when students come and go, also demands that the institution owns research data. On the other hand, universities seldom monetize their data. Companies, on the other hand, recognize the strategic value of the data they amass (even if not directly monetized), particularly those in the information technology sector.

Universities create policies both for *Data Security* and *Data Sharing* (also called open science). Security policies tend to be all-encompassing for all types of data, not just research, categorizing data based on source, use and privacy considerations. Data security policy aims to comply with regulations (particularly data subject HIPAA and FERPA), and to ensure the authenticity, confidentiality and continuity of essential data. On the other hand, data sharing—often freely to anyone—is essential to many areas of research, and may even be required by sponsors to aid in replication and research progress. Yet even when data is freely shared, it must also be protected against alteration or destruction. Naturally, sharing under open science policies can conflict with industry sponsor needs for protecting proprietary data, and must be resolved as agreements are negotiated.

Last, both universities and companies care about *Publicity* and therefore include provisions regulating how each other's name or trademarks are used. Neither would want the other to publicly suggest that an agreement or a project implies an endorsement, or for their name to be used in advertising without explicit permission. Some universities will explicitly prohibit any implied endorsement of a company or its products. On the other hand, universities will typically not agree to secrecy over the existence of an agreement. This principle of transparency was explicit in the Pajaro Dunes Statement (Unauthored, 1982), prepared by a group of elite research universities, which was issued soon after the Bayh–Dole Act was enacted. Public disclosure of agreements may also be required under state laws.

14.3.3 *Culture and Leadership*

As a UI relationship transitions from execution of an agreement to execution of the work stated in the agreement, companies and universities must ensure they will meet their commitments. By design, UI relationships blend organizations with aligned goals and distinct cultures. The value of the relationship comes from synergizing complementary cultures, but so do the risks. Participants on both sides need to break habits that do not fit the relationship. Administration and leadership must

recognize that a UI engagement is not the same as federally sponsored research, consulting or pure service.

A path to success is adopting principles outlined in Chap. 3. As an institution, universities must articulate the strategy, values, and mission that places UI relationships within their overarching mission of research, learning, and public service. They must also state how their policies, agreements, units, and stakeholders can contribute to and support UI engagement, including those policies that define how people are appraised, rewarded, and promoted. Last, they must convey how their attitudes, capabilities, services, and attributes align with a comprehensive engagement strategy. In each of these areas, they must assess whether relationships with industry are supported and how they will use such relationships to enhance their reputation for innovation and impact. Without such support, achieving a culture through which UI engagement can thrive will be impossible.

Likewise, companies must make similar assessments and develop a unified strategy that supports UI relationships. As in universities, such a strategy must be comprehensive, because many people within companies impact UI agreements, and can either nurture or inhibit such relationships. UI engagement cannot be viewed as a series of one-off agreements, but part of a strategy that will help the company succeed through new technology, creative insights, and talent recruitment.

14.4 Takeaways and Future Strategies

UI relationships begin when companies and universities discover that their goals are aligned and their capabilities complement each other. Yet even when both sides see value in a relationship, execution of formalized engagements through agreements is challenged by differences in organizational values and policies and by compliance with laws and regulations.

Success comes from careful consideration of how the details of agreements—the terms and conditions—align with the purpose of the relationship. Specific details to resolve include

- What information and results should be transparent and open to all and what must be protected for confidentiality or privacy?
- How will research be conducted, to give confidence in the quality of the results?
- What are the potential conflicts of interest, and how will they be managed?
- What laws and regulations apply to the relationship, and how will compliance be assured?
- How will IP be managed and protected, and what is the pathway through which the company can commercialize research results?

For these questions and others, negotiators must understand how atypical terms and agreements can catalyze risk-informed success, within the context of complementary organizations. The following chapters drill deeper into key mechanisms for

success: IP management, data access generation and use, and international partnerships.

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Randolph Hall is Dean's Professor of Industrial and Systems Engineering and Director of CREATE at the University of Southern California. His experience includes serving for nearly 15 years as USC's VP of Research, where he oversaw research advancement, administration, and ethics for the entire university. He is the author of *Managing Innovation Inside Universities* and *Queueing Methods for Services and Manufacturing*, and Editor of the *Handbook of Transportation Science, Patient Flow*, and the *Handbook of Healthcare System Scheduling*. He has written numerous articles on how universities can innovate in their practices and previously served as board chair for the UIDP. Hall holds a BS in Industrial Engineering and Operations Research and a PhD in Civil Engineering from UC Berkeley.